Linear Regression in Machine Learning

What is Linear Regression?

Linear Regression is one of the most fundamental and widely used algorithms in machine learning.
It is a **supervised learning algorithm** used for **predicting a continuous output** (like house prices, temperature, or stock prices) based on one or more input features.

In simple words:

Linear Regression finds the best straight line (or plane) that fits the data points.

This line (or plane) can then be used to **predict** future values.

Key Idea

The goal of Linear Regression is to find a **linear relationship** between the input variables (**X**) and the output variable (**Y**).

Mathematically, for a **simple linear regression** (one input feature):

$$y = mx + c$$

Where:

- y = predicted output
- x = input feature
- m = slope of the line (also called weight or coefficient)
- c = y-intercept (bias)

Example:
Predict a student's exam score (`y`) based on the number of study hours (`x`).

Types of Linear Regression

Type	Description
Simple Linear Regression	One independent variable, one dependent variable.
Multiple Linear Regression	Multiple independent variables predicting one dependent variable.
Polynomial Regression	Extension where the relationship is not strictly linear but still modeled as a polynomial (still solved like linear regression but nonlinear in nature).

Visual Representation

Imagine plotting points on a graph:

- X-axis: number of study hours
- Y-axis: exam score

Linear Regression will draw the **best straight line** through the points that minimizes the error between the points and the line.

How Does Linear Regression Work?

1. **Start with a dataset** of (X, Y) values.
2. **Assume a straight-line model** y=mx+c
3. **Find the best m and c** by minimizing the difference between actual values and predicted values.
4. This difference is measured by **Loss Functions**, usually:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where:

- yi = actual value
- y^i = predicted value
- n = number of samples

5. **Use optimization algorithms** like **Gradient Descent** to find the best m and c that minimize the error.

Important Concepts

1. Cost Function

- A function that measures how far off the model's predictions are from the actual results.
- In Linear Regression, usually the Mean Squared Error (MSE).

2. Gradient Descent

- An optimization technique to **adjust m and c** gradually to minimize the cost function.

3. Overfitting and Underfitting

- **Overfitting**: Model too complex; fits noise.
- **Underfitting**: Model too simple; cannot capture trends.
- Linear Regression is **prone to underfitting** if relationships are non-linear.

Advantages of Linear Regression

- Simple to understand and implement.
- Computationally fast.
- Works well for problems where the relationship between variables is linear.

Limitations of Linear Regression

- Assumes linear relationship between inputs and output.
- Sensitive to outliers.
- Poor performance if there is multicollinearity (features highly correlated with each other).
- Doesn't handle complex patterns (like images or natural language).

Evaluation Metrics for Linear Regression

Why Do We Need Evaluation Metrics?

When we build a Linear Regression model, it’s not enough to just "predict."
We need to **measure how good or bad the model is** at making predictions.

Evaluation metrics help us:

- Quantify how close the predictions are to the actual values.
- Compare different models.
- Understand if a model is underfitting, overfitting, or performing well.

Common Evaluation Metrics for Linear Regression

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1. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

- Measures the **average absolute difference** between actual and predicted values.
- **Easy to understand**: it's just the average of the absolute errors.
- **Lower MAE** = Better Model.

Example: If MAE = 5, it means on average, your model is off by 5 units.

2. Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- Measures the **average of the squared differences** between actual and predicted values.
- **Squaring** penalizes larger errors more than smaller ones.
- **Lower MSE** = Better Model.

Tip: MSE is sensitive to outliers.

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

- RMSE is just the **square root of MSE**.
- It brings the error back to the **same unit** as the original output.
- **Lower RMSE** = Better Model.

Interpretation: If RMSE = 10, on average your prediction is about 10 units off.

4. R-squared (R^2 Score)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Where:

- SSres = Sum of squares of residuals (errors)
- SStot = Total sum of squares
- **R^2 measures how much of the variation** in the output can be explained by the model.
- **Range:**

$$0 \leq R^2 \leq 1$$

- **Higher R^2** = Better model.

Example:
If R^2=0.85, it means **85%** of the variance in the target variable is explained by the model.

Quick Comparison Table

Metric	What It Measures	Good Value	Notes
MAE	Average absolute error	Lower is better	Simple but treats all errors equally
MSE	Average squared error	Lower is better	Penalizes big errors more
RMSE	Root of MSE	Lower is better	Same units as original output
R^2 Score	Variance explained by the model	Closer to 1 is better	Good for overall model fit